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★ REVIEWS

Hospitalization cost of conventional psychiatric care compared to broad-spectrum micronutrient treatment: literature review and case study of adult psychosis

Background: Healthcare costs are skyrocketing, with mental health treatment amongst the most expensive, especially when hospitalization is involved. According to the Mental Health Commission of Canada, one in five Canadians is living with a mental disorder in any given year, at an annual cost of $50 billion. In light of this societal burden, alternative approaches are being evaluated, such as brief psychotherapy by phone, peer support, and, as part of the emerging field of nutritional mental health, treatment with micronutrients (minerals and vitamins). Effectiveness of micronutrients has been demonstrated for many types of psychiatric symptoms, in about 45 studies of formulas that are either multinutrient (e.g., several B vitamins) or broad-spectrum (usually over 20 minerals and vitamins). Although this literature demonstrates therapeutic benefits, the potential economic impact of micronutrient treatment has been evaluated in only one case study of childhood psychosis.

Methods: The current case study was initiated to evaluate mental health-related hospitalization costs from 1997 to 2003 for a female adult diagnosed with various mood and psychotic symptoms. She was treated for the first 5 years with conventional methods and then subsequently with a broad-spectrum micronutrient formula.[†]

Results: The patient's annual mental health hospitalization costs during conventional treatment averaged $59,864 across 5 years (1997–2001), with a peak annual cost of about $140,000. Since transitioning to broad-spectrum micronutrients, she has incurred no provincial hospitalization costs for mental health care, though her self-funded costs are currently $720/year for the micronutrients.

Conclusion: Further exploration of the treatment of mental health problems with broad-spectrum micronutrient formulas has the potential to make two significant contributions: improved mental health, and decreased costs for governments.

Hospitalization cost of conventional psychiatric care compared to broad-spectrum micronutrient treatment: literature review and case study of adult psychosis

Kaplan BJ, Isaranuwatchai W, Hoch JS. Int J Ment Health Syst. 2017 Jan 31;11:14.

FULL TEXT

Vitamin–mineral treatment of attention-deficit hyperactivity disorder in adults: double-blind randomised placebo-controlled trial.

A micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy, showed "statistically robust improvements" in the first-ever double-blind study to test the effectiveness and safety of a multivitamin-mineral supplement in adults with ADHD. Researchers from the University of Canterbury and the University of Otago, New Zealand, randomized 80 adults diagnosed with ADHD to take either the micronutrient formulation or identical-looking placebo pills for 8 weeks. Most of the study participants had at least one psychiatric diagnosis in addition to ADHD. Additional diagnoses included multiple anxiety disorders, major depressive disorder, dysthymia, bipolar disorder, reading disability, and alcohol/substance misuse or dependence. The nutrient group reported more than double the improvement in attention, hyperactivity, and impulsivity symptoms, compared with the placebo group. Clinical psychologists rated more than twice as many people in the nutrient group 'very much' or 'much' improved in overall symptoms. They also rated moderate and severely depressed participants in the nutrient group as having nearly double the improvement in depression symptoms. Researchers found that the micronutrients were completely safe; there were no differences in side effects between the two groups.

Vitamin–mineral treatment of attention-deficit hyperactivity disorder in adults: double-blind randomised placebo-controlled trial.

Rucklidge JJ, Frampton CMA, Gorman B, Boggis A. The British Journal of Psychiatry. 2014 Feb;204(2): doi: 10.1192/bjp.bp.113.132126 [Epub ahead of print]

FULL TEXT

Feasibility of a nutritional supplement as treatment for pediatric bipolar spectrum disorders.

Researchers investigated the therapeutic effects of a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy in 7 children with bipolar disorder. The children's additional psychiatric diagnoses included attention deficit hyperactivity disorder, oppositional defiant disorder, major depressive disorder, generalized anxiety disorder, conduct disorder, and obsessive-compulsive disorder. The children experienced a 37% decrease in depression scores and a 45% decrease in mania scores over 8 weeks. Side effects were minor and transient—mostly temporary gastric discomfort.

Feasibility of a nutritional supplement as treatment for pediatric bipolar spectrum disorders.

Frazier EA, Fristad MA, Arnold LE. Journal of Alternative and Complementary Medicine. 2012 Jul;18(7):678-85.

FULL TEXT

Can micronutrients improve neurocognitive functioning in adults with ADHD and severe mood dysregulation? A pilot study.

Researchers studied the impact of a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy on neurocognitive functioning in 14 adults with attention deficit hyperactivity disorder (ADHD) and severe mood dysregulation over an 8-week period. Significant improvement was observed in the ADHD-micronutrient group, but not the control group, across a range of verbal abilities, including verbal learning, verbal cognitive flexibility and fluency, and verbal inhibition. These neurocognitive improvements were large and consistent with improved psychiatric functioning.

Can micronutrients improve neurocognitive functioning in adults with ADHD and severe mood dysregulation? A pilot study.

Rucklidge JJ, Harrison R, Johnstone J. Journal of Alternative and Complementary Medicine. 2011 Dec;17(12):1125-31.

FULL TEXT

Database analysis: Micronutrient formulation greatly improves bipolar and ADHD symptoms in 161 children and adolescents.

Scientists analyzed data from 120 children with pediatric bipolar disorder who were treated with a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy for at least 6 months. 24% of these children also had a diagnosis of attention deficit hyperactivity disorder (ADHD). Their results were compared with an additional 41 children who were diagnosed with ADHD alone. Children with a bipolar diagnosis showed a 59% decline in symptoms, whereas children with ADHD showed a 40% decrease in symptoms. The duration of symptom reduction suggests that benefits of micronutrient treatment were not attributable to placebo or expectancy effects. Of those taking psychiatric medications, more than half were able to completely discontinue them over a 6-month period. Medication use by the remainder of children was reduced by 74% during the same time period.

Database analysis of children and adolescents with bipolar disorder consuming a micronutrient formula.

Rucklidge JJ, Gately D, Kaplan BJ. BioMed Central Psychiatry. 2010 Sep 28;10:74.

FULL TEXT

Successful treatment of bipolar disorder II and ADHD with a micronutrient formula: a case study.

Researchers documented the effects of a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy on psychiatric and neurocognitive functioning in a naturalistic off-on-off-on design. A 21-year old female with bipolar II disorder, attention deficit hyperactivity disorder (ADHD), social anxiety, and panic disorder began taking a vitamin-mineral formulation following a documented 8-year history of ongoing psychiatric symptoms that were not well managed by medications. After 8 weeks on the formula, she showed significant improvements in mood, anxiety, and hyperactivity/impulsivity. She then chose to come off the formula. After 8 weeks, her depression scores returned to pre-treatment levels, and anxiety and ADHD symptoms worsened dramatically. When the formula was reintroduced, all psychiatric symptoms showed gradual improvements. After one year, she was in remission of all mental illness diagnoses. Her neurocognitive changes mirrored behavioral changes, showing improved processing speed, variability in response, and verbal memory.

Successful treatment of bipolar disorder II and ADHD with a micronutrient formula: a case study.

Rucklidge JJ, Harrison R. CNS Spectrums. 2010 May;15(5):289-95.

FULL TEXT

Database analysis: Micronutrient formulation greatly improves bipolar symptoms in 358 adults.

Scientists analyzed data from 358 adults with a diagnosis of bipolar disorder who took a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy for 6 months or more. The adults' overall symptom severity was 41% lower than baseline after 3 months, and 45% lower after 6 months. In addition, at 6 months, 53% of adults experienced greater than 50% improvement in symptoms, with 32% of adults experiencing greater than 75% improvement in symptoms. Symptom improvements at 6 months suggested that benefits of micronutrient treatment were not attributable to placebo or expectancy effects. Of those taking psychiatric medications, nearly half were able to discontinue them over a 6-month period. The remainder reduced their medication usage by 54% during the same time period. Adults who gradually eliminated their medications and took full recommended levels of the vitamin-mineral formulation experienced the greatest improvements in symptoms.

Database analysis of adults with bipolar disorder consuming a micronutrient formula.

Gately D, Kaplan BJ. Clinical Medicine Insights: Psychiatry. 2009 Apr;4:3-16.

FULL TEXT

Multinutrient supplement as treatment: literature review and case report of a 12-year-old boy with bipolar disorder.

Researchers describe a 12-year-old boy who was diagnosed at age 6 with bipolar disorder. His diagnosis evolved to include psychotic features, generalized anxiety disorder and obsessive-compulsive disorder. Although he was treated with psychiatric medications for 6 years, he continued to experience many debilitating mood, anxiety, and obsessive-compulsive symptoms. During treatment with a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy, the boy's global functioning notably improved. He began interacting more appropriately with peers, remained calm and playful throughout most of the day, slept throughout the night, remained focused and efficient while completing schoolwork, and experienced decreased compulsions. Eventually, his hallucinations also ceased and all diagnoses fully remitted. His symptoms remained stable during a 14-month follow-up period.

Multinutrient supplement as treatment: literature review and case report of a 12-year-old boy with bipolar disorder.

Frazier EA, Fristad MA, Arnold LE. Journal of Child and Adolescent Psychopharmacology. 2009 Aug;19(4):453-60.

FULL TEXT

Micronutrient formulation greatly improves mood, behavior, and anxiety disorders in 9 children.

Researchers treated nine unselected children with mood and behavioral problems with a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy. The children's psychiatric diagnoses included attention deficit hyperactivity disorder, bipolar disorder, depression, oppositional defiant disorder, obsessive-compulsive disorder, generalized anxiety disorder, Asperger syndrome, and Prader-Willi syndrome. After a minimum of 8 weeks of micronutrient treatment, the children's behavior improved significantly in the following areas: anxious and depressed mood, thought problems, interpersonal relationship problems, attention problems, social problems, withdrawn behavior, disruptive behavior, delinquent behavior, aggressive behavior, and self-harm behavior.

Improved mood and behavior during treatment with a mineral-vitamin supplement: an open-label case series of children.

Kaplan BJ, Fisher JE, Crawford SG, Field CJ, Kolb B. Journal of Child and Adolescent Psychopharmacology. 2004 Spring;14(1):115-22.

FULL TEXT

Micronutrient formulation greatly improves bipolar symptoms in 19 adults.

Dr. Miles Simmons, a psychiatrist in private practice, followed nineteen patients diagnosed with bipolar I or bipolar II disorder for an average of 13 months while they took a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy. Sixteen of the 19 patients (84%) showed significant clinical improvement. Thirteen of 16 medicated patients were able to completely discontinue psychiatric medications over 3 to 10 weeks and maintain stability over an average follow-up period of 13 months.

Nutritional approach to bipolar disorder.

Simmons M. The Journal of Clinical Psychiatry. 2003 Mar;64(3):338; author reply 338-9.

FULL TEXT

Micronutrient formulation greatly improves bipolar symptoms in 23 adults, adolescents, and children.

After observing a 10-year-old boy's dramatic response to a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy, Dr. Charles Popper, a psychiatrist and psycho-pharmacologist at Harvard Medical School, treated 22 additional bipolar disorder-diagnosed patients in his private practice (10 adults, 9 adolescents, and 3 preadolescents) with the micronutrients. Nineteen of the 22 patients (86%) showed a positive response to micronutrient treatment. Of the 15 patients being treated with medications when they began the nutritional supplement, 11 patients remained stable during a follow-up period of 6 to 9 months without psychiatric medications.

Do vitamins or minerals (apart from lithium) have mood-stabilizing effects?

Popper CW. The Journal of Clinical Psychiatry. 2001 Dec;62(12):933-5.

FULL TEXT

Micronutrient formulation stabilizes mood in 11 bipolar adults.

Researchers treated 11 adults diagnosed with various types of bipolar disorder for an average of 44 weeks with a micronutrient formulation co-developed[†] by Hardy Nutritionals® founder David Hardy. The adults' additional diagnoses included attention deficit hyperactivity disorder, dysthymic disorder, and obsessive-compulsive disorder. The adults experienced a 55% reduction in depression symptoms, a 60% reduction in mania symptoms, and a 66% reduction in general psychiatric symptoms. Their psychiatric medication use decreased by more than 50%. The only reported side effect, nausea, was infrequent, minor, and transitory.

Effective mood stabilization with a chelated mineral supplement: an open-label trial in bipolar disorder.

Kaplan BJ, Simpson JS, Ferre RC, Gorman CP, McMullen DM, Crawford SG. The Journal of Clinical Psychiatry. 2001 Dec;62(12):936-44.

FULL TEXT

[†] The micronutrient formulation studied was a pre-2013 version of *Truehope EMPowerplus* which was co-formulated by David Hardy and Anthony Stephan. *Truehope EMPowerplus* is a registered trademark of The Synergy Group of Canada Inc., which was co-founded by David Hardy and Anthony Stephan in 1999. David Hardy officially resigned as a shareholder of The Synergy Group of Canada Inc. and director of Truehope Nutritional Support Ltd. in 2013 to focus his efforts exclusively on Hardy Nutritionals®.

STUDIES MENU

- 1-year naturalistic follow-up of a Randomised Double-Blind, Placebo-Controlled Trial ("NoMAD") Exploring the Effectiveness of Micronutrients in Improving Symptoms of Anxiety and Depression
- Gut microbiome changes with micronutrient supplementation in children with attention-deficit/hyperactivity disorder: the MADDY study
- Potential link between copy number variation and abnormal genome wide DNA methylation profile in an individual with severe ADHD and a strong response to micronutrient treatment
- Investigation of a broad-spectrum micronutrient formulation as a possible precipitant of pharmacokinetic micronutrient-drug interactions
- Broad-Spectrum Micronutrients or Antidepressants for Antenatal Depression: Effect on Maternal and Infant Birth Outcomes in an Observational Secondary Analysis of NUTRIMUM
- Efficacy and safety of a mineral and vitamin treatment on symptoms of antenatal depression: 12-week fully blinded randomised placebo-controlled trial (NUTRIMUM)
- Problems most concerning to parents of children with ADHD and emotional dysregulation in a randomized controlled trial of multinutrients: MADDY secondary analysis
- Treatment response to supplemental nutrients for ADHD is independent of diet quality: the MADDY Study RCT
- Micronutrients for ADHD in youth (MADDY) study: comparison of results from RCT and open label extension.
- Exploring the impact of antenatal micronutrients used as a treatment for maternal depression on infant temperament in the first year of life
- Household Food Insecurity Is Associated with Symptoms of Emotional Dysregulation in Children with Attention Deficit Hyperactivity Disorder: The MADDY Study.
- Effect of antenatal micronutrient or antidepressant exposure on Brazelton neonatal behavioral assessment scale (NBAS) performance within one-month of birth
- Efficacy and safety of a vitamin-mineral intervention for symptoms of anxiety and depression in adults: A randomised placebo-controlled trial "NoMAD"
- Non-pharmacological interventions for attention-deficit hyperactivity disorder in children and adolescents
- Inflammation and Vitamin C in Women with Prenatal Depression and Anxiety: Effect of Multinutrient Supplementation.
- Diet Quality as a Moderator of Response to Multinutrients for ADHD and Emotional Dysregulation: The MADDY RCT.
- Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youth: A Placebo-Controlled Randomized Clinical Trial
- Paediatric adverse event rating scale: a measure of safety or efficacy? Novel analysis from the MADDY study
- Clinician guidelines for the treatment of psychiatric disorders with nutraceuticals and phytoceuticals: The World Federation of Societies of Biological Psychiatry (WFSBP) and Canadian Network for Mood and Anxiety Treatments (CANMAT) Taskforce
- Parents' priorities and preferences for treatment of children with ADHD: Qualitative inquiry in the MADDY study
- Do Changes in Blood Nutrient Levels Mediate Treatment Response in Children and Adults With ADHD Consuming a Vitamin-Mineral Supplement?
- Multinutrients for the Treatment of Psychiatric Symptoms in Clinical Samples: A Systematic Review and Meta-Analysis of Randomized Controlled Trials
- A Holistic Self-learning Approach for Young Adult Depression and Anxiety Compared to Medication-Based Treatment-As-Usual
- Resting-state networks and neurometabolites in children with ADHD after 10 weeks of treatment with micronutrients: results of a randomised placebo-controlled trial
- Novel Mineral–Vitamin Treatment for Reduction in Cigarette Smoking: A Fully Blinded Randomized Placebo-Controlled Trial
- Human gut microbiome changes during a 10 week Randomised Control Trial for micronutrient supplementation in children with attention deficit hyperactivity disorder
- Mineral-Vitamin Treatment Associated with Remission in Attention-Deficit/Hyperactivity Disorder Symptoms and Related Problems: 1-Year Naturalistic Outcomes of a 10-Week Randomized Placebo-Controlled Trial
- Multinutrients for ADHD Youth (MADDY) Trial: Preliminary Trends for Treatment
- An Observational Preliminary Study on the Safety of Long-Term Consumption of Micronutrients for the Treatment of Psychiatric Symptoms
- Can we predict treatment response in children with ADHD to a vitamin-mineral supplement? An investigation into pre-treatment nutrient serum levels, MTHFR status, clinical correlates and demographic variables
- Methylomic changes in response to micronutrient supplementation and MTHFR genotype.
- Hospitalization cost of conventional psychiatric care compared to broad-spectrum micronutrient treatment: literature review and case study of adult psychosis
- Vitamin-mineral treatment improves aggression and emotional regulation in children with ADHD: a fully blinded, randomized, placebo-controlled trial.
- Vitamin–Mineral Treatment of ADHD in Adults: A 1-Year Naturalistic Follow-Up of a Randomized Controlled Trial
- Micronutrient Treatment of Emotional Dyscontrol Following Traumatic Brain Injury
- Effect of Micronutrients on Insomnia in Adults: A Multiple-Baseline Study
- A randomised trial of nutrient supplements to minimise psychological stress after a natural disaster
- Clinically Significant Symptom Reduction in Children with Attention-Deficit/Hyperactivity Disorder Treated with Micronutrients: An Open-Label Reversal Design Study
- Vitamin–mineral treatment of attention-deficit hyperactivity disorder in adults: double-blind randomised placebo-controlled trial.
- Moderators of treatment response in adults with ADHD treated with a vitamin-mineral supplement
- Psychological functioning 1 year after a brief intervention using micronutrients to treat stress and anxiety related to the 2011 Christchurch earthquakes: a naturalistic follow-up
- Nutritional and Safety Outcomes from an Open-Label Micronutrient Intervention for Pediatric Bipolar Spectrum Disorders
- Could yeast infections impair recovery from mental illness? A case study using micronutrients and olive leaf extract for the treatment of ADHD and depression.
- Use of micronutrients attenuates cannabis and nicotine abuse as evidenced from a reversal design: a case study.
- Efficacy and cost of micronutrient treatment of childhood psychosis.
- Micronutrient formulation outperforms Berocca Performance in reducing depression, anxiety, and stress in 91 earthquake survivors.
- Feasibility of a nutritional supplement as treatment for pediatric bipolar spectrum disorders.
- Safety and tolerability of Micronutrient formulations.
- Can micronutrients improve neurocognitive functioning in adults with ADHD and severe mood dysregulation? A pilot study.
- Micronutrient formulation reduces depression, anxiety, and stress in 17 earthquake survivors with ADHD.
- Micronutrients reduce stress and anxiety following a 7.1 earthquake in adults with Attention-Deficit/Hyperactivity Disorder
- Effect of micronutrients on behaviour and mood in adults with ADHD: Evidence from an 8-week open label trial with natural extension.
- Database analysis: Micronutrient formulation greatly improves bipolar and ADHD symptoms in 161 children and adolescents.
- Successful treatment of bipolar disorder II and ADHD with a micronutrient formula: a case study.
- Micronutrient formulation outperforms medications in 88 patients with autism.
- Micronutrient formulation greatly improves brain function in rats.
- Database analysis: Micronutrient formulation greatly improves bipolar symptoms in 358 adults.
- Successful treatment of OCD with a micronutrient formula following partial response to Cognitive Behavioral Therapy (CBT): a case study.
- Multinutrient supplement as treatment: literature review and case report of a 12-year-old boy with bipolar disorder.
- Micronutrient formulation greatly improves mood, behavior, and anxiety disorders in 9 children.
- Treatment of mood lability and explosive rage with minerals and vitamins: two case studies in children.
- Micronutrient formulation greatly improves bipolar symptoms in 19 adults.
- Micronutrient formulation greatly improves bipolar symptoms in 23 adults, adolescents, and children.
- Micronutrient formulation stabilizes mood in 11 bipolar adults.

Consult with your doctor when altering your treatment regimen and before adjusting your medication dosing.

Information from Hardy Nutritionals® is for educational purposes only, and suggestions should not be considered medical advice.

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